

Quiz 3

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Put today's date here

First, please enter the name of each group member in the AUTHOR field at the top! As you work in this R project, I will ask you to copy and paste some of your work over into your group's shared quiz document. You can find the link to the shared document in Blackboard.

You may need the `tidyverse`, `ggdist`, and `ggribes` packages in your work. Start by loading them all.

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(ggdist)
library(ggribes)

##
## Attaching package: 'ggribes'
##
## The following objects are masked from 'package:ggdist':
##
##   scale_point_color_continuous, scale_point_color_discrete,
##   scale_point_colour_continuous, scale_point_colour_discrete,
##   scale_point_fill_continuous, scale_point_fill_discrete,
##   scale_point_size_continuous
```

Loading and examining the data

Load the data file called `nba_players_19.csv` from our textbook's web site. Here is the web address for the dataset: https://www.openintro.org/data/csv/nba_players_19.csv We store the data file under the name `players`.

```
players <- read.csv("https://www.openintro.org/data/csv/nba_players_19.csv")
```

You will find the codebook for the data at

https://www.openintro.org/data/index.php?data=nba_players_19

Explore the dataset in the usual ways – view the dataset, read the codebook.

```
#head and summary of the df
summary(players)
```

```
##   first_name      last_name      team      team_abbr
## Length:494      Length:494      Length:494      Length:494
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##   position      number      height
## Length:494      Min.   : 0.00   Min.   :69.00
## Class :character 1st Qu.: 6.00   1st Qu.:77.00
## Mode  :character Median :13.00   Median :79.00
##                  Mean   :17.51   Mean   :79.05
##                  3rd Qu.:25.00   3rd Qu.:82.00
##                  Max.   :99.00   Max.   :87.00
##                  NA's    :1
```

```
head(players,10)
```

```
##   first_name last_name team team_abbr position number height
## 1      Alex   Abrines Thunder   OKC      Guard      8      78
## 2     Jaylen   Adams   Hawks   ATL      Guard     10      74
## 3     Steven   Adams   Thunder  OKC      Center     12      84
## 4        Bam   Adebayo   Heat    MIA Center-Forward 13      82
## 5   DeVaughn Akoon-Purcell Nuggets  DEN  Guard-Forward 23      78
## 6   LaMarcus Aldridge   Spurs    SAS      Forward 12      83
## 7     Rawle   Alkins   Bulls    CHI      Guard 20      77
## 8    Grayson   Allen   Jazz     UTA      Guard 24      77
## 9    Jarrett   Allen   Nets     BKN      Center 31      83
## 10 Al-Farouq   Aminu   Blazers   POR      Forward  8      81
```

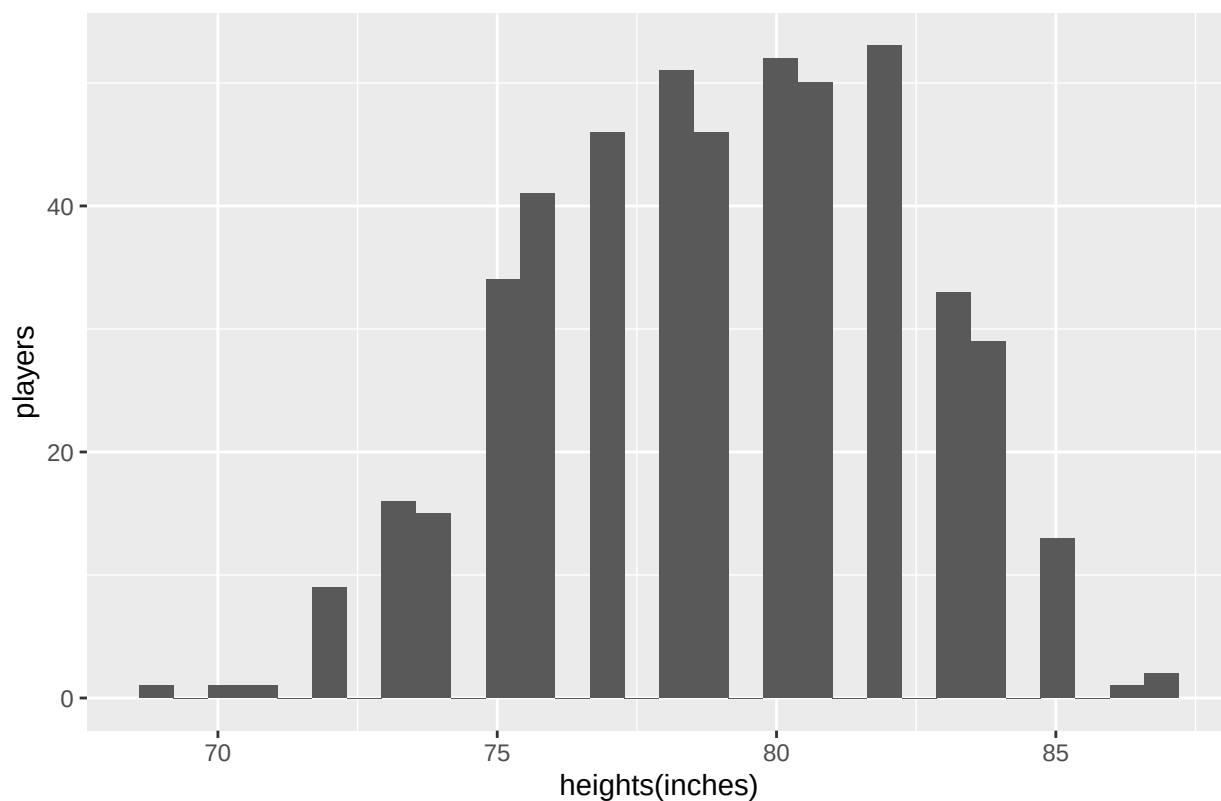
1. Make an appropriate graph showing the distribution of players' heights in this dataset. You will be describing the distribution next, so make several graphs and consider which one you think will lend itself best to the distribution. Think about this variable and make sure to look at it in the dataframe. Is it categorical or quantitative? What sorts of graphs can you use?

There are many possibilities here!

```
ggplot(data = players, aes(x = height)) +
  geom_histogram() +
  labs(title = "distribution of players' heights", x = "heights(inches)", y = "players")
```

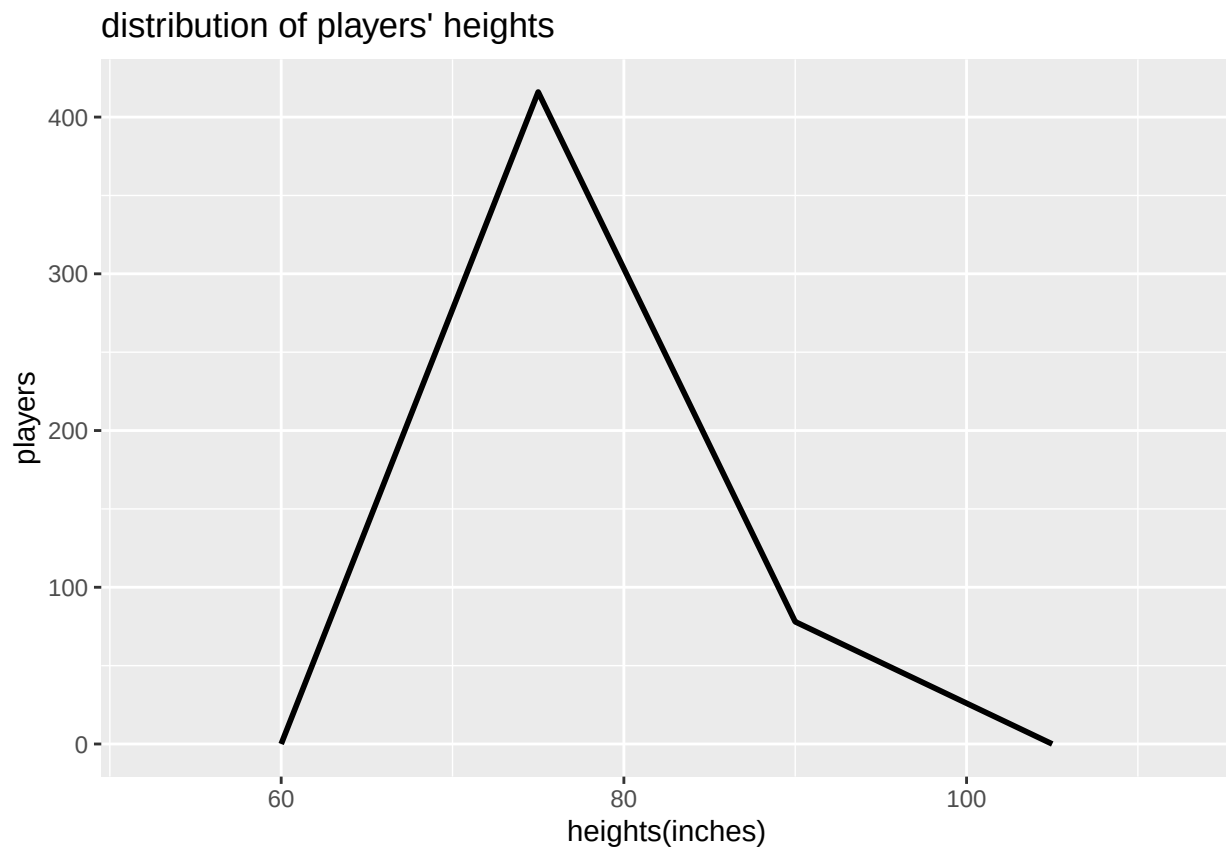
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

distribution of players' heights



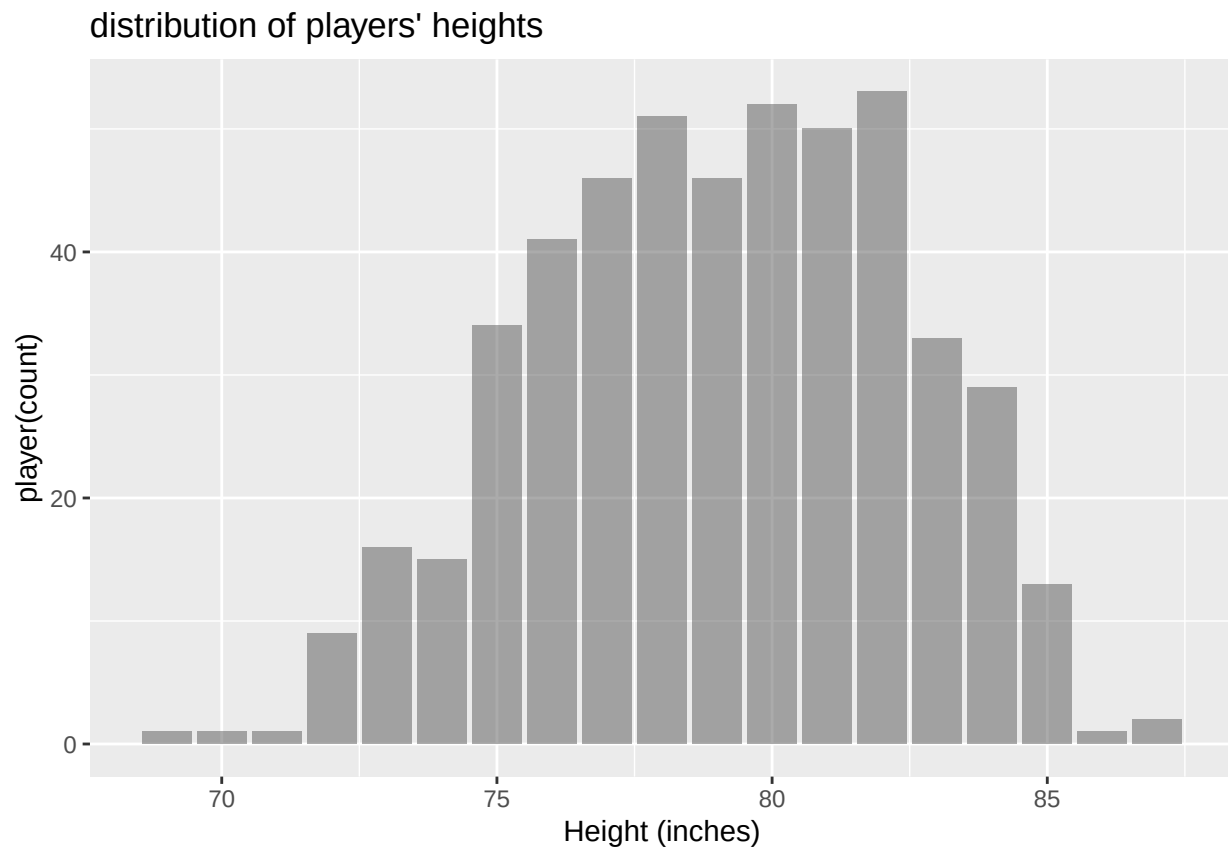
```
ggplot(players, aes(x = height)) +  
  geom_freqpoly(binwidth = 15, size = 1) +  
  labs(title = "distribution of players' heights", x = "heights(inches)", y = "players")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use `linewidth` instead.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



```
ggplot(data = players, mapping = aes(x = height )) +  
  geom_bar(shape = 1, alpha = 0.5) +  
  labs(  
    title = "distribution of players' heights",  
    x = "Height (inches)",  
    y = "player(count)"  
  )
```

```
## Warning in geom_bar(shape = 1, alpha = 0.5): Ignoring unknown parameters:  
## `shape`
```



When you are done, PASTE a screenshot of your favorite graph on your group's shared document.

in Gdocs

2. Calculate some summary statistics for the players' heights including the mean, standard deviation, median, and the quantiles we have been calculating.

```
#mean, median, sd
mean_height <- mean(players$height, na.rm = TRUE)
median_height <- median(players$height, na.rm = TRUE)
sd_height <- sd(players$height, na.rm = TRUE)
```

```
mean_height
```

```
## [1] 79.05263
```

```
median_height
```

```
## [1] 79
```

```
sd_height
```

```
## [1] 3.31284
```

```
#quantiles
quantile_results <- players %>%
  summarize(
    q025 = quantile(height, 0.025, na.rm = TRUE),
```

```
q25 = quantile(height, 0.25, na.rm = TRUE),
q75 = quantile(height, 0.75, na.rm = TRUE),
q975 = quantile(height, 0.975, na.rm = TRUE))
quantile_results
```

```
##    q025 q25 q75 q975
## 1    73  77  82   85
```

Paste a screenshot of your summary statistics onto the shared document.

in Gdoc

3. NOW GO TO QUESTION 3 ON YOUR GROUP'S SHARED DOCUMENT TO WRITE A DESCRIPTION OF THIS GRAPH.

in Gdoc

Comparing positions

In the remaining questions, you will compare the heights of players in different positions. There are some minor combined positions (such as Center-Forward), which we will ignore. The following command will make a new data frame (called `players_filtered`) for you with only players having one of the three most-common positions (Center, Forward, Guard).

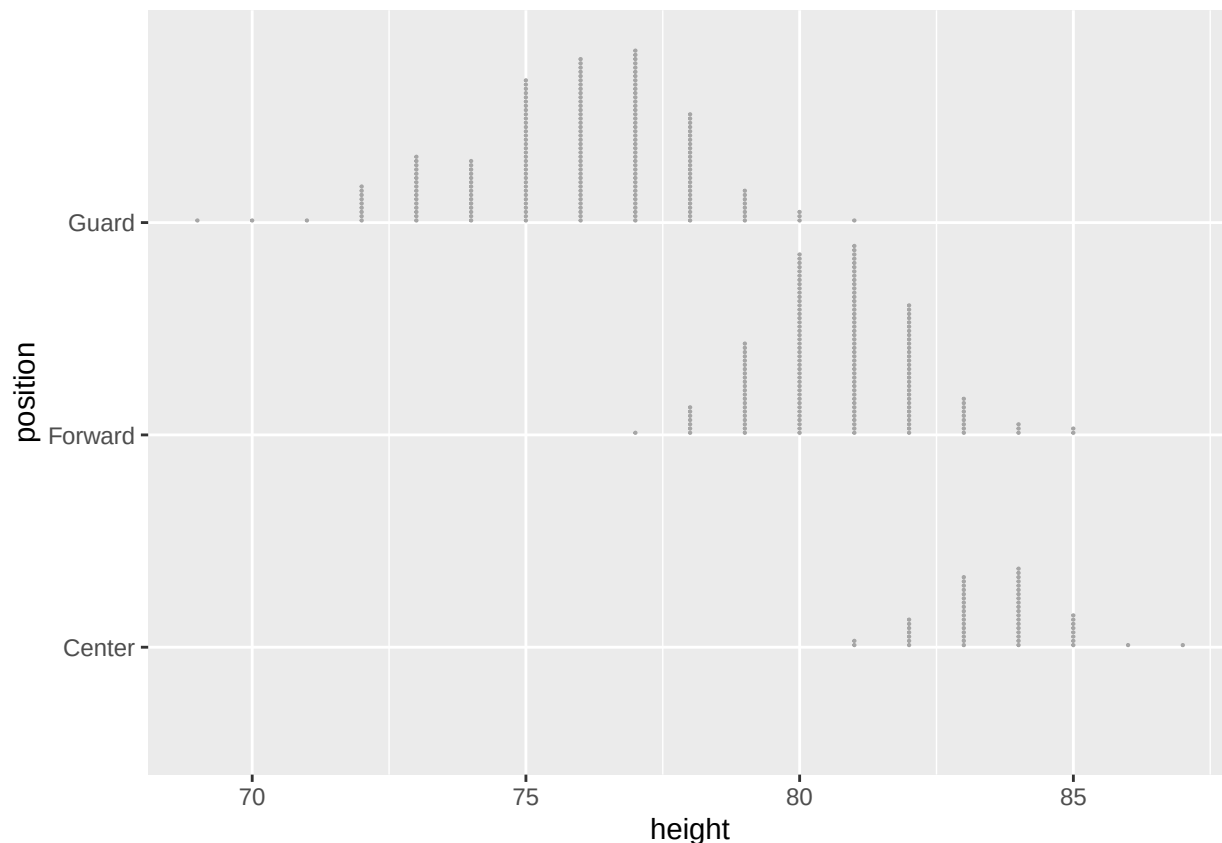
Click **PLAY** on this code chunk. Don't forget this as it is important moving forwards!

```
players_filtered <- players %>%
  filter(position %in% c("Center", "Forward", "Guard"))
```

4. Make a graph showing the distribution of heights grouped by position. (You may want to experiment, once again, to find a nice-looking graph to describe.)

There are again several reasonable approaches.

```
ggplot(data = players_filtered, mapping = aes(x = height, y = position)) +
  geom_dots()
```



PASTE YOUR GRAPH INTO THE SHARED DOCUMENT.

5. Calculate some useful summary statistics for the heights grouped by position.

```
# Calculate mean, median, and standard deviation for the entire dataset
mean_by_position <- mean(players$height, na.rm = TRUE)
median_by_position <- median(players$height, na.rm = TRUE)
sd_by_position <- sd(players$height, na.rm = TRUE)
```

```
# results
```

```
mean_by_position
```

```
## [1] 79.05263
```

```
median_by_position
```

```
## [1] 79
```

```
sd_by_position
```

```
## [1] 3.31284
```

```
# Calculate summary statistics for heights grouped by position
quantile_results2 <- players_filtered %>%
  group_by(position) %>%
  summarize(
    q025 = quantile(height, 0.025, na.rm = TRUE),
    q25 = quantile(height, 0.25, na.rm = TRUE),
    q50 = quantile(height, 0.50, na.rm = TRUE),
```

```

q75 = quantile(height, 0.75, na.rm = TRUE),
q975 = quantile(height, 0.975, na.rm = TRUE),
mean_height = mean(height, na.rm = TRUE),
median_height = median(height, na.rm = TRUE),
sd_height = sd(height, na.rm = TRUE)
)

# quantile results
quantile_results2

## # A tibble: 3 x 9
##   position q025  q25  q50  q75  q975 mean_height median_height sd_height
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl>      <dbl>          <int>      <dbl>
## 1 Center    81.4   83   84   84  85.6      83.6            84       1.17
## 2 Forward   78    80   81   82  83.9      80.7            81       1.41
## 3 Guard    72    75   76   77  79       75.8            76       2.00

```

6. GO TO THE SHARED SLIDES to write a description of this graph.

In Gdoc

KNIT your file and submit it. Also, download the shared word/google document as a PDF and upload it for grading.